1 Core content

1.1 The impact of new and emerging technologies

1.1.1 Industry

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- Modern factories increasingly use **automated production**. Robots are often used to complete some repetitive and monotonous tasks previously carried out by humans, which can lead to unemployment.
- **Computer-aided manufacture (CAM)** machines are used where high volumes of identical products of a consistent high quality are needed.
- People often move to different countries to find suitable employment or where there is a shortage in a particular workforce. This is referred to as **demographic movement**.
- Science and technology parks are areas where like-minded businesses come together to form centres of excellence.

Automated production: a production method of using machinery controlled by computers.

CAM: computer-aided manufacture.

1.1.2 Enterprise

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- A privately-owned business is owned by the company founders, or their families, or by a small group of investors whose shares in the company are not publicly traded.
- Crowd funding is an internet-based scheme allowing people to raise money to get a business idea started, such as the manufacture of a new product. Investors lend money to projects that they believe are viable.
- The UK government offers some loans and grants to new start-up businesses. Investors can get tax relief on their investments. While not all new businesses are successful, the potential to boost the economy and the provision of future employment is worth taking a risk.
- Not-for-profit organisations reinvest all profits back into the business for the benefit of all involved.



Figure 1.1 The Forest Stewardship Council® (FSC®) is an example of a not-for-profit organisation

1.1.3 Sustainability

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- **Sustainability** is about meeting today's needs without compromising the needs of future generations.
- Demand on the world's natural resources currently exceeds supply. Materials used in product design fall into two categories:
  - **finite**: resources with a limited supply that will eventually run out
  - **non-finite**: resources that can be replenished and are unlikely to run out.
- One way to reduce our environmental impact is to consider the six Rs.

Sustainability: the manufacture of goods without compromising future needs; the materials required for a product can be replenished and will not run out.

Table 1.1 The six Rs of sustainability

Rethink	Is there a better way of making the product that is less harmful to the environment? Can the design be simplified to make manufacturing easier?
Recycle	Can the product be recycled easily after it is no longer needed? Can the components and materials be separated easily? Can recycled materials be used?
Repair	Can this product be repaired easily if it breaks? Can the component parts be replaced easily?
Refuse	Consumers might not buy a product if it is not environmentally friendly or if it is made by people working in poor conditions.
Reduce	Can the number of component parts, new materials or packaging be reduced? Can the manufacturing process be simplified to reduce the energy used?
Reuse	Can any parts be reused after it is no longer needed? Could the product be reused for something else once its primary use is no longer required?

Transportation

- Each product has a **carbon footprint** from its manufacture and the distance it travels until it is purchased and used.
- Transportation costs/carbon footprint can be reduced by:
  - sourcing materials closer to manufacturing centres and potential markets
  - using energy-efficient means of transporting goods, for example electric or **hybrid vehicles**.

Carbon footprint: a measure of the greenhouse gases emitted by human activity.

Hybrid vehicles: vehicles that are both petrol and electric, so emissions are reduced.

Pollution

The manufacture and use of any product leads to an increase in pollution levels. There are ways to reduce the level of pollution.

- Manufacturers could use cleaner energy sources, such as solar or wind.
- Goods could be transported using electric vehicles that do not emit greenhouse gases.
- Products could be built to last and easy to repair, causing less waste.
- Products need to be disposed of carefully if they cannot be recycled, avoiding landfill where possible.



Figure 1.2 Hybrid technology reduces carbon emission from vehicles

Demand on natural resources

- Natural resources and productive land enable us to produce the goods and services that sustain and support modern lifestyles.
- The **ecological footprint** is a measure of the impact that human activity has on the environment.
- Humanity's ecological footprint is currently 1.7 Earths – the Earth takes about 18 months to replace what we use in 12 months. If this continues, we will create an **ecological deficit**.

Ecological footprint: a measure of the impact that human activity has on the environment.

Ecological deficit: when resources are being used up faster than nature can replenish them.